



RV College of Engineering®

Mysore Road, RV Vidyaniketan Post, Bengaluru - 560059, Karnataka, India

Go, change the world

Automated Speech Recognition for Dysarthria

An Interdisciplinary Project Report (CS367P)

Submitted by,

Unmesh Raj

1RV23AI129

Vijesh

1RV23IS137

Sanjay C M

1RV23ET039

Seshasai Chillara

1RV23ET040

Under the guidance of

Dr. Shanta Rangaswamy

Professor & Head

Dept. of CSE

RV College of Engineering®

In partial fulfillment of the requirements for the degree of
Bachelor of Engineering in respective departments

2025-26



RV College of Engineering®, Bengaluru
(Autonomous institution affiliated to VTU, Belagavi)



CERTIFICATE

Certified that the interdisciplinary project (CS367P) work titled ***Automated Speech Recognition for Dysarthria*** is carried out by _____ of RV College of Engineering, Bengaluru, in partial fulfillment of the requirements for the degree of **Bachelor of Engineering** in respective departments during the year 2025-26. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the interdisciplinary project report deposited in the departmental library. The interdisciplinary project report has been approved as it satisfies the academic requirements in respect of interdisciplinary project work prescribed by the institution for the said degree.

Dr. Shanta Rangaswamy
Guide

Dr. Shanta Rangaswamy
Head of the Department

Dr. M.V. Renukadevi
Dean Academics

Dr. K. N. Subramanya
Principal

External Viva

Name of Examiners

Signature with Date

- 1.
- 2.

DECLARATION

We, **Unmesh Raj, Vijesh, Sanjay C M, Seshasai Chillara** students of sixth semester B.E., RV College of Engineering, Bengaluru, hereby declare that the interdisciplinary project titled '**Automated Speech Recognition for Dysarthria**' has been carried out by us and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering** in respective departments during the year 2025-26.

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any degree or diploma to any other university.

We also declare that any Intellectual Property Rights generated out of this project carried out at RVCE will be the property of RV College of Engineering, Bengaluru and We will be one of the authors of the same.

Place: Bengaluru

Date:

Name

Signature

1. Unmesh Raj(1RV23AI129)
2. Vijesh(1RV23IS137)
3. Sanjay C M(1RV23ET039)
4. Seshasai Chillara(1RV23ET040)

ACKNOWLEDGEMENTS

We are indebted to our guide, **Dr. Shanta Rangaswamy**, Professor & Head, Department of CSE, RVCE for the wholehearted support, suggestions and invaluable advice throughout our project work and also helped in the preparation of this thesis.

We also express our gratitude to our panel member **Dr. Prapulla S B**, Associate Professor, Department of CSE, RVCE for the valuable comments and suggestions during the phase evaluations.

Our sincere thanks to the project coordinator **Dr. Prapulla S B**, Associate Professor, Department of CSE for timely instructions and support in coordinating the project.

Our gratitude to **Prof. Narashimaraja P**, Department of ECE, RVCE for the organized latex template which made report writing easy and interesting.

Our sincere thanks to **Dr. Shanta Rangaswamy**, Professor and Head, Department of Computer Science and Engineering, RVCE for the support and encouragement.

We are deeply grateful to **Dr. M.V. Renukadevi**, Professor and Dean Academics, RVCE, for the continuous support in the successful execution of this Interdisciplinary project.

We express sincere gratitude to our beloved Professor and Vice Principal, **Dr. Geetha K S**, RVCE and Principal, **Dr. K. N. Subramanya**, RVCE for the appreciation towards this project work.

Lastly, we take this opportunity to thank our family members and friends who provided all the backup support throughout the project work.

ABSTRACT

Dysarthria is a motor speech disorder caused by neurological conditions such as ALS, Cerebral Palsy, Parkinsons disease, and stroke. It affects the muscles used for speech production, leading to slurred or slow speech that can be difficult to understand. The severity of dysarthria can vary widely, and it can significantly impact an individual's ability to communicate effectively. Traditional methods of assessing and treating dysarthria often involve in-person evaluations and therapy sessions, which can be time-consuming and may not be accessible to all patients. Globally, millions of people suffer from speech impairments, yet most speech technologies are trained only on normal speech. Therefore, there is a need for innovative approaches that can provide more efficient and accessible solutions for individuals with dysarthria. This report explores the development of a novel system designed to address these challenges by leveraging advanced technologies to improve the daily life of dysarthric patients.

The objectives of this work is to generate synthetic dysarthric speech using Text-to-Speech (TTS) and GAN-based voice conversion techniques and to develop a post-ASR semantic correction layer utilizing large language models for improved transcription accuracy which enhances system robustness through noise augmentation and advanced acoustic feature extraction methods. We finally wish to integrate a Personal Knowledge Graph to support contextual speech disambiguation

We were able to achieve a significant improvement in the accuracy of dysarthric speech recognition by implementing a novel architecture that combines advanced acoustic feature extraction methods with a post-ASR semantic correction in the layers of whisper model by openai. The system was trained and evaluated using a dataset of dysarthric speech samples, and we observed a sub 20% WER in transcription accuracy which is better compared to previous approaches. The integration of a Personal Knowledge Graph for contextual speech disambiguation further enhanced the system's performance, particularly in noisy environments.

CONTENTS

Abstract	i
List of Figures	iv
List of Tables	v
1 A Hybrid Approach for Dysarthric Speech Recognition and Correction	1
1.1 Introduction	2
1.2 Literature Review	2
1.3 Motivation	3
1.4 Problem Statement	3
1.5 Objectives	3
1.6 Brief Methodology of the Project	4
1.7 Assumptions and Constraints	4
1.8 Organization of the Report	5
2 Theory and Fundamentals of Analog to Digital converter	6
2.1 Contents of this chapter	7
2.2 Contents of this chapter	7
2.3 Contents of this chapter	7
2.4 Contents of this chapter	7
2.5 Use of Acronyms and Glossaries	7
3 Analysis/Design of Pipelined Analog to Digital converter	9
3.1 Contents of this Chapter	10
3.2 Paraphrasing	10
3.3 Quotations	11
4 Implementation of Pipelined Analog to Digital converter	12
4.1 Contents of this chapter	13
5 Results & Discussions	14
5.1 Contents of this chapter	15

5.2	Tables in thesis	15
5.3	Math equation in thesis	16
5.4	Interpretation of results	17
6	Conclusion and Future Scope	18
6.1	Conclusion	19
6.2	Future Scope	19
6.3	Learning Outcomes of the Project	19
A	Code	20
A.1	First Appendix	21



LIST OF FIGURES



LIST OF TABLES

5.1 Country List	16
----------------------------	----





Chapter 1

A Hybrid Approach for Dysarthric Speech Recognition and Correction

CHAPTER 1

A HYBRID APPROACH FOR DYSARTHRIC SPEECH RECOGNITION AND CORRECTION

1.1 Introduction

Voice has become the default interface for technology, yet for millions of people, it remains unusable. Dysarthria, a motor speech disorder caused by neurological conditions such as stroke, ALS, and cerebral palsy, severely impacts articulation and intelligibility. According to Census 2011 estimates, nearly 7% of the Indian population lives with some form of disability, with a significant portion facing speech-related impairments. Globally, tens of millions are affected, yet modern speech recognition systems continue to overlook this group.

While state-of-the-art Automatic Speech Recognition (ASR) systems such as Whisper achieve Word Error Rates (WER) below 10% for normal speech, their performance deteriorates drastically to 30–70% for dysarthric speech. This gap is not merely technical but deeply social, limiting accessibility to digital systems, communication tools, and assistive technologies. The challenge lies in the variability of speech patterns, lack of large-scale datasets, and sensitivity to environmental noise.

This project attempts to bridge this gap by combining data augmentation, synthetic speech generation, and semantic correction. Instead of relying solely on acoustic perfection, the system shifts focus toward understanding intent, enabling meaningful communication even when speech signals are degraded.

1.2 Literature Review

Recent research has explored multiple directions to improve dysarthric speech recognition. Shrestha et al. proposed a two-stage data augmentation pipeline, improving ASR performance by combining static and dynamic augmentation techniques. However, their approach struggles to generalize across patients due to high variability in speech patterns.

Vij et al. introduced a system combining CNN, Whisper, and T5 models for cerebral palsy speech detection, achieving improved classification but still suffering from low transcription accuracy. Zhang et al. proposed personalized style embeddings, enabling adaptation to individual speech patterns, but requiring sufficient speaker-specific data.

Patel et al. leveraged Mel spectrograms and transfer learning, showing improved performance, though dependent on dataset similarity. Huang et al. developed a voice conversion framework capable of generating dysarthric speech, but struggled with preserving speaker identity.

More recent approaches integrate language models. Mehta et al. demonstrated an LLM-based communication system that corrects dysarthric speech outputs, highlighting the importance of semantic understanding. Huang et al. proposed rhythm-based synthesis (DARS), improving modeling of speech patterns, though computationally intensive.

Shahamiri addressed phoneme distortions using visual spectrogram-based augmentation, achieving significant improvements for severe dysarthria. Kumar et al. generated synthetic dysarthric speech using fine-tuned TTS models, though such data often fails to capture real-world variability.

Overall, existing works focus heavily on acoustic modeling, with limited emphasis on contextual correction. Dataset scarcity, speaker variability, and lack of real-world robustness remain unresolved challenges.

1.3 Motivation

Despite rapid advancements in AI, accessibility for speech-impaired individuals remains limited. High ASR error rates prevent effective communication, restricting independence and digital participation. The absence of diverse datasets and personalized models further amplifies this issue.

This project is motivated by the need to build a system that not only recognizes speech but understands intent. By integrating synthetic data, noise robustness, and semantic reasoning, the system aims to provide a practical assistive solution.

1.4 Problem Statement

Current Automatic Speech Recognition systems perform poorly on dysarthric speech due to limited datasets, acoustic variability, and environmental noise. This results in high transcription error rates and limited accessibility of voice-based technologies for individuals with speech impairments.

1.5 Objectives

The objectives of the project are:

1. To generate synthetic dysarthric speech using Text-to-Speech (TTS) and GAN-based voice conversion techniques
2. To develop a post-ASR semantic correction layer utilizing large language models for improved transcription accuracy
3. To enhance system robustness through noise augmentation and advanced acoustic feature extraction methods
4. To integrate a Personal Knowledge Graph to support contextual speech disambiguation

1.6 Brief Methodology of the Project®

The proposed system follows a hybrid pipeline combining acoustic modeling and semantic reasoning. First, synthetic dysarthric speech is generated using Matcha-TTS and StarGAN-VC, increasing dataset diversity. This data is further augmented with environmental noise to simulate real-world conditions.

A frozen Whisper-v3-Turbo encoder extracts robust acoustic features from input speech. These features are mapped into the embedding space of a local large language model using a trainable projection layer. The LLM then acts as a semantic repair engine, correcting transcription errors using contextual reasoning and a Personal Knowledge Graph built from user-specific data.

This approach shifts the paradigm from pure transcription to intent-aware communication.

1.7 Assumptions and Constraints

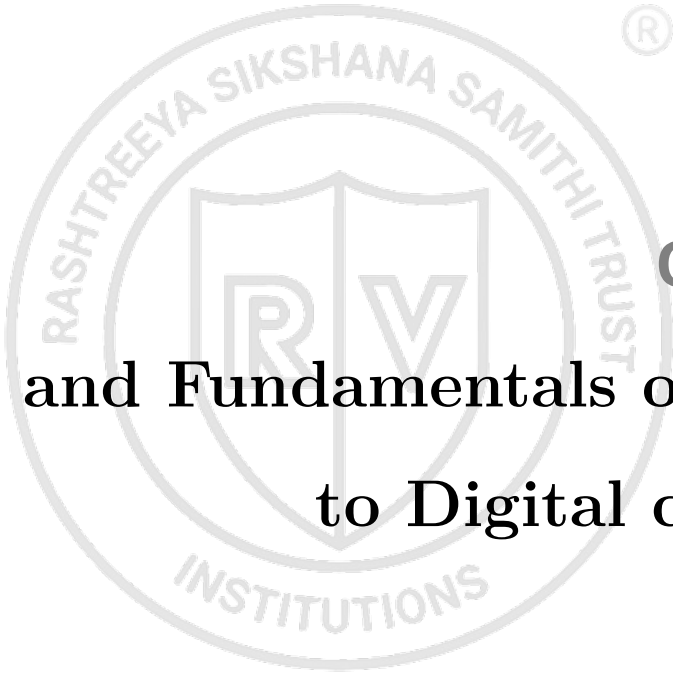
- Availability of large-scale dysarthric speech datasets is limited
- Synthetic speech may not fully replicate real-world speech impairments
- Model performance depends on variability across speakers and conditions
- Computational constraints limit full-scale training of large models
- System evaluation is constrained by dataset diversity and noise conditions

1.8 Organization of the Report

This report is organized as follows:

- Chapter 2 discusses the fundamental concepts of speech recognition and neural models
- Chapter 3 presents the system architecture and design methodology
- Chapter 4 details the implementation and experimental setup
- Chapter 5 presents results and performance evaluation using WER and CER metrics
- Chapter 6 concludes the work and outlines future research directions





Chapter 2

Theory and Fundamentals of Analog to Digital converter

CHAPTER 2

THEORY AND FUNDAMENTALS OF ANALOG TO DIGITAL CONVERTER

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters 2 to end with "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

2.1 Contents of this chapter

This chapter should discuss about the prerequisite learnings before the execution of the project. Organise and elaborate the theory and necessary fundamentals required for the execution of the project. You can use \subsections and subsubsections in this chapter.

2.2 Contents of this chapter

If a specific programming language is required for the project, a section can be allotted in this chapter to discuss it.

2.3 Contents of this chapter

Tools used could be another possible section to discuss about the software tools used in the work.

2.4 Contents of this chapter

The details in this chapter can be added in consultation with the project guide. For an internship based projects, subsections can be modified accordingly.

2.5 Use of Acronyms and Glossaries

Acronyms are nothing but the short form of regular repeated word. Say for example, you have a repeat word "Integrated Circuits" and you want to use a short form for it as "IC". For which you have to first define the word and use it wherever you wanted to refer it.

First, let's look at the definition, which has to be entered in `Glossaries.tex` under `CoverPages` directory.

```
%\newacronym{<Ref>}{<Short-Form>}{<Expanded word>}
\newacronym{ic}{IC}{Integrated Circuits}
```

In order to use the defined acronym, use the commands `\gls{<Ref>}` as shown below

As an example, call the definition with `\gls{ic}` and the outcome of it is reflected as, Integrated Circuit (IC).

Note: For the First time, the expanded form appears along with the Short-form definition inside parenthesis. But when the `\gls{}` is repeated, only Short-form appears inside the parenthesis.

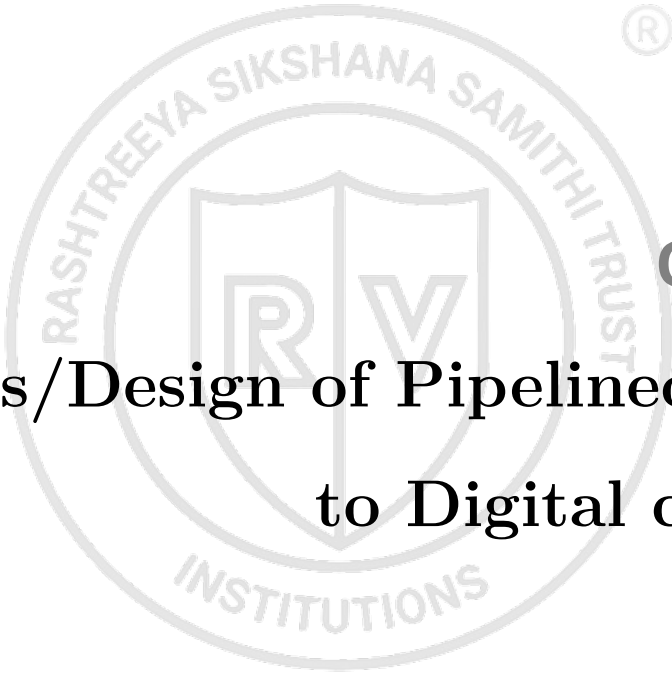
Now, let's look at the definition of symbols. Follow the syntax to define the symbol first, inside `Glossaries.tex` under `CoverPages` directory.

```
%\newglossaryentry{<Ref>}{name=<Symbol>, description={<description about the symbol>}}
\newglossaryentry{rc}{name=$\tau$, description={Time constant}, type=symbolList}
```

As an example, the rate of change is defined with `\gls{rc}` and the outcome of it is reflected as, the rate of change is defined with τ .

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.



Chapter 3

Analysis/Design of Pipelined Analog to Digital converter

CHAPTER 3

ANALYSIS/DESIGN OF PIPELINED ANALOG TO DIGITAL CONVERTER

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

3.1 Contents of this Chapter

This chapter should contain the following sections and subsections in detail.

1. Specifications for the Design
2. Pre analysis work for the design or Models used
3. Design methodology in detail
4. Design Equations
5. Experimental techniques (if any)

Apart from the aforementioned sections, you can add sections as per the requirements of the project in consultation with your guide.

3.2 Paraphrasing

When you paraphrase a written passage, you rewrite it to state the essential ideas in your own words. Because you do not quote your source word for word when paraphrasing, it is unnecessary to enclose the paraphrased material in quotation marks. However, the paraphrased material must be properly referenced because the ideas are taken from someone else whether or not the words are identical.

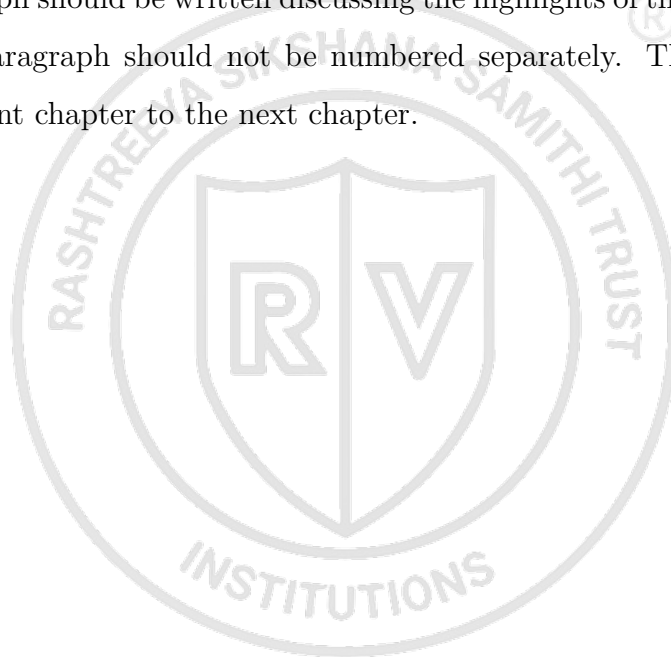
Ordinarily, the majority of the notes you take during the research phase of writing your report will paraphrase the original material. Paraphrase only the essential ideas. Strive to put original ideas into your own words without distorting them."

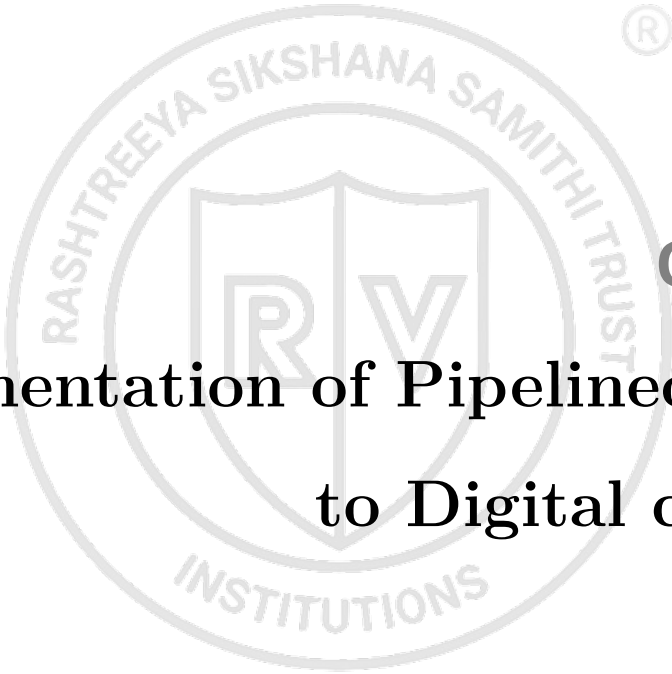
3.3 Quotations

When you have borrowed words, facts, or idea of any kind from someone else's work, acknowledge your debt by giving your source credit in footnote (or in running text as cited reference). Otherwise, you will be guilty of plagiarism. Also, be sure you have represented the original material honestly and accurately. Direct word to word quotations are enclosed in quotation marks.”

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.





Chapter 4

Implementation of Pipelined Analog to Digital converter

CHAPTER 4

IMPLEMENTATION OF PIPELINED ANALOG TO DIGITAL CONVERTER

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

4.1 Contents of this chapter

This chapter should elaborate the following in detail.

1. Implementation details for hardware based projects
2. Top level Design for software based projects

You can add sections and sub sections to elaborate your project work done.

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.



Chapter 5

Results & Discussions

CHAPTER 5

RESULTS & DISCUSSIONS

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

5.1 Contents of this chapter

All the results obtained for your objectives should be discussed in this chapter. This chapter should contain the following sections as per the project.

1. Simulation results
2. Experimental results
3. Performance Comparison
4. Inferences drawn from the results obtained

All the figures should be properly explained by bringing the scenarios of the design done in the project. A detailed discussion of results obtained should be done in this chapter.

5.2 Tables in thesis

- All Table Caption should be in Sentence Case, TNR 10 Pt. It should be of the Format:
 - Table 1.1 Results of the experiment .(Centered)
- It should be cited as Table 1.1.
- Caption should appear above the Table.
- Table Header and the entries should be of Font TNR 10 Pt, Justified.
- For wider Table, the page orientation can be Landscape.

- For Larger Table, it can run to pages and the header should be repeated for each page of the Table.
- Table must be adjusted to fit in the page and no single row is left out for a new page.

Sample Table 5.1 is given below for your reference,

Table 5.1: Country List

Country Name or Area Name	ISO ALPHA 2 Code	ISO ALPHA 3 Code	ISO numeric Code
Afghanistan	AF	AFG	004
Aland Islands	AX	ALA	248
Albania	AL	ALB	008
Algeria	DZ	DZA	012
American Samoa	AS	ASM	016
Andorra	AD	AND	020
Angola	AO	AGO	024

5.3 Math equation in thesis

All equation should be written using equation editor or using an equivalent tool.

- Equations should be numbered as : 1.1, 1.2 ...
- Equation should be Centered, 12 Pt, TNR.
- Equation number should be right Justified
- It should be cited as Eqn. 1.1.
- If the sentence starts by citing an equation, then it should be written as Equation 1.1 For example, Equation 5.1 states the Pythagoras theorem.

For example in Eqn. 5.1, The well known Pythagorean theorem $x^2 + y^2 = z^2$ was proved to be invalid for other exponents. Meaning the next equation has no integer solutions:

$$x^n + y^n = z^n \quad (5.1)$$

The mass-energy equivalence is described by the famous equation in Eqn. 5.2

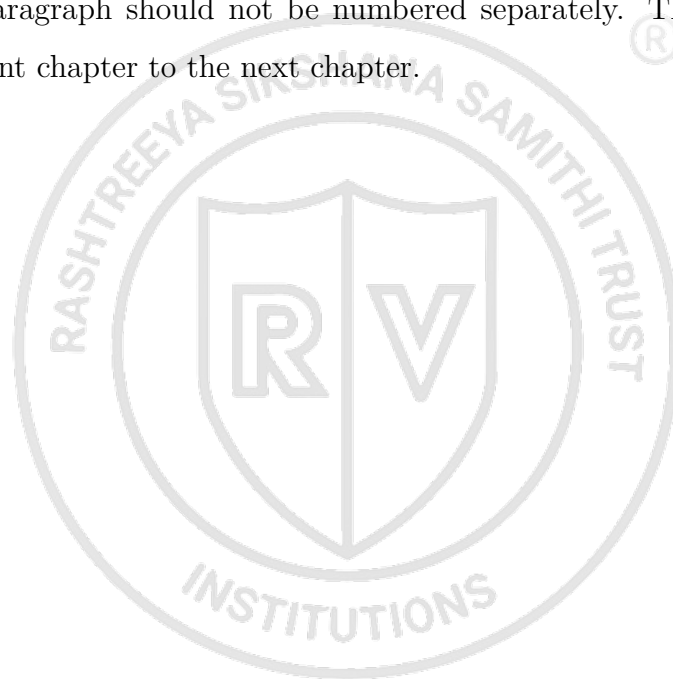
$$E = mc^2 \quad (5.2)$$

discovered in 1905 by Albert Einstein.

5.4 Interpretation of results

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.





Chapter 6

Conclusion and Future Scope

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

6.1 Conclusion

This chapter should not contain an introduction paragraph like other chapters. You can directly write conclusion of the work done under this section. Typically this section can have 3 to 4 paragraphs.

First paragraph should bring in the scenario of the project and every objective should be explained here.

Second paragraph should say how the objectives are implemented and achieved.

Last paragraph should draw the conclusions from each objective with quantitative results, performance improvement etc.

6.2 Future Scope

Briefly discuss the constraints and limitations of the project and state the possibilities of extending the work in future.

6.3 Learning Outcomes of the Project

- List the learning outcomes here
- List a minimum of 5 learning outcomes



Appendix A

Code

APPENDIX A

CODE

A.1 First Appendix

You can use `tcbllisting` for creating the code snippets. The following example illustrates how one can customize the `tcbllisting` to achieve the `tcl` script. Similarly, one can use it for other programming language listing, including HDL.

```
# Since our design has a clock with name clk,
## specify that name under [get_port ]
create_clock -period 40 -waveform {0 20} [get_ports clk]

# Setting a 'delay' on the clock:
set_clock_latency 0.3 clk

# Setting up constraints on your I/P and O/P pins
set_input_delay 2.0 -clock clk [all_inputs]
set_output_delay 1.65 -clock clk [all_outputs]

# Set realistic 'loads' on each output pin
set_load 0.1 [all_outputs]

# Set 'maximum' fanin and fan-out for the input and output pins
set_max_fanout 1 [all_inputs]
set_fanout_load 8 [all_outputs]
```